

# Lucky Verma

MACHINE LEARNING ENGINEER | AGENTIC SYSTEMS | KNOWLEDGE GRAPHS | RAG

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## SUMMARY

Machine Learning Engineer with 4+ years building production agentic AI systems, knowledge graphs, and deep learning pipelines for large-scale data infrastructure. Public research record includes two arXiv preprints with code/artifacts. Expert in multi-agent orchestration (LangGraph, DSPy), RAG pipelines, entity resolution, model evaluation, and real-time inference serving on GPU clusters.

## TECHNICAL SKILLS

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| AI / LLM  | RAG, Agentic AI, <b>LangGraph</b> , <b>DSPy</b> , LangChain, vLLM, Fine-tuning (LoRA, RLHF/DPO), Tool Use, HF Transformers                                 |
| ML / DL   | <b>PyTorch</b> , TensorFlow, FSDP, DeepSpeed, SLURM Multi-GPU, Triton Kernels, <b>Deep Learning</b> , <b>Computer Vision</b> , NLP, <b>CUDA</b>            |
| Infra     | <b>Knowledge Graphs</b> (Apache AGE, LightRAG), PostgreSQL/pgvector, <b>FastAPI</b> , <b>Docker</b> , K8s, Terraform, <b>AWS</b> , Redis, Model Monitoring |
| Languages | <b>Python</b> , SQL, C++, JavaScript, Bash                                                                                                                 |

## WORK EXPERIENCE

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| <b>Eccalon LLC (AI and federal data systems)</b><br><b>MACHINE LEARNING ENGINEER</b> <ul style="list-style-type: none"><li>Architected and deployed production <b>deep learning</b> + agentic AI workflows for regulated document analysis, evidence retrieval, and corporate ownership reasoning using <b>LangGraph</b> orchestration, <b>DSPy</b> optimization, and multi-step reasoning across federal and corporate data sources.</li><li>Engineered <b>knowledge graph</b> infrastructure (PostgreSQL + Apache AGE) with entity resolution across SAM.gov, SEC EDGAR, and GLEIF corporate registries, supporting multi-hop ownership-chain tracing with low-latency query response.</li><li>Built production <b>document intelligence pipeline</b> with GPU-accelerated embeddings (<b>Ollama</b> + NIM fallback), LightRAG/PostgreSQL-backed retrieval, and source-grounded synthesis across federal databases and regulatory documents.</li><li>Built <b>model evaluation</b> framework for LLM baselines, feature engineering, contamination checks, and multi-annotator validation in ownership and compliance-oriented reasoning workflows.</li><li>Deployed production <b>RAG pipelines</b> on AWS with multi-path reasoning, hybrid vector/keyword search, and source-grounded retrieval; built GPU-accelerated CV pipelines with <b>PyTorch</b> and <b>CUDA</b>.</li><li>Owned end-to-end <b>ML infrastructure</b> — Docker multi-service orchestration, Terraform IaC, <b>CI/CD</b>, <b>model serving</b>, <b>monitoring</b>, and multi-backend embedding fallback for production ML services.</li></ul> | Nashville, TN<br>Jun 2022 – Present           |
| <b>University of Maryland, Baltimore County</b><br><b>GRADUATE RESEARCH ASSISTANT</b> <ul style="list-style-type: none"><li>Developed ensemble <b>1D U-Net</b> for biomedical signal classification (PPG → sleep staging), achieving <b>92%</b> accuracy; built web-based data science learning platform using CV/ML pipelines with <b>pandas</b> and <b>Streamlit</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Baltimore, MD<br>Jan 2022 – May 2023          |
| <b>Vast Dream Group</b><br><b>AI SPECIALIST</b> <ul style="list-style-type: none"><li>Built recommendation engine using <b>BART</b> zero-shot classification (96% accuracy on MNLI); architected AI measurement systems for optics industry, contributing to <b>\$1M+</b> revenue pipeline across 3 product lines.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Mumbai, India (Remote)<br>Jan 2021 – Aug 2021 |
| <b>TandM Techlabs</b><br><b>SOFTWARE DEVELOPER</b> <ul style="list-style-type: none"><li>Built urban mobility platform with geospatial ML: <b>Pandana</b> accessibility metrics, <b>Valhalla/OSMnx</b> route optimization on <b>PostGIS</b>, cached with <b>Redis</b> and containerized on <b>Kubernetes</b> — serving real-time discovery for B2B logistics.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Secunderabad, India<br>Jun 2018 – Dec 2020    |

## RESEARCH & OPEN SOURCE

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| <b>When Does Removing LayerNorm Help?</b><br><i>Activation bounding   Transformer normalization   Reproducible artifacts</i> <ul style="list-style-type: none"><li>Solo-author public preprint studying regime-dependent behavior when replacing LayerNorm with bounded activations; released configs, result manifests, validation scripts, and artifact notes.</li></ul>   | arXiv   Code |
| <b>When Top-1 Fails: Calibrating LoRA Monitors for Masked Diffusion LMs</b><br><i>Masked diffusion LMs   LoRA monitor calibration   Released aggregate artifacts</i> <ul style="list-style-type: none"><li>Co-authored public preprint with reference logging/checking scripts and sanitized aggregate result artifacts backing paper tables and figures.</li></ul>          | arXiv   Code |
| <b>Weight Decay Regimes in Grokking Transformers</b><br><i>Grokking diagnostics   Transformer training behavior   Public dataset</i> <ul style="list-style-type: none"><li>Solo-author public preprint with diagnostics code, aggregate JSONs, provenance manifests, Lean 4 checks, and a Hugging Face dataset.</li></ul>                                                    | arXiv   Code |
| <b>Public ML and developer-tool repositories</b><br><i>97 public repos   50 non-fork public repos   126 total stars as of Jun 2026</i> <ul style="list-style-type: none"><li>Selected public work spans LayoutLM document classification (35 stars), driver-license OCR (26), PPG/EEG biomedical signal projects, MWB Linux, Mail Reclaim, and InfiniteContext-1B.</li></ul> | GitHub       |

## EDUCATION

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| <b>University of Maryland, Baltimore County</b><br><i>Master of Science in Computer Science – GPA: 3.97</i> <ul style="list-style-type: none"><li><b>DARPA Catalyst Award</b> (\$50K) – Caselet, selected from <b>250+ teams</b> for AI-powered national security workforce tools</li></ul> | Baltimore, MD<br>May 2023  |
| <b>SRM University</b><br><i>Bachelor of Technology in Electrical &amp; Electronics Engineering – GPA: 3.9</i>                                                                                                                                                                               | Chennai, India<br>May 2019 |